

SEC P vs NP log 2026-05-19

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-19 00:46 UTC

SEC P vs NP — daily log 2026-05-19

5 cycles recorded

Decision Tree Depth and Communication Complexity via Lifting

- **Verdict:** INCONCLUSIVE
- **Bridge:** `DECISION_TREE_DEPTH` × `COMMUNICATION_COMPLEXITY`
- **Recorded:** 2026-05-19 00:11 UTC
- **Entry ID:** eeec3f948dcf

Statement

For any Boolean function f on n variables, if the decision tree depth of f is d , then the deterministic communication complexity $R_{\{1/3\}}(f)$ is $\Omega(d / \log n)$.

Rationale

This conjecture explores the relationship between the depth of a decision tree and the communication complexity of the function, leveraging lifting techniques to establish a lower bound. If true, it would imply that functions with deeper decision trees require more communication, which is a key insight in understanding the interplay between these complexity measures.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1302.4207v1] A composition theorem for decision tree complexity - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [0911.3482v5] Complexity of Networks (reprise) - [0101006v1] On Complexity and Emergence

Empirical Test

- exit code: 124, elapsed: 240.11s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode 124) without producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce the parameter grid (e.g., restrict to $n \in \{8, 16, 24\}$ and sample fewer depths) and/or optimize the deterministic CC computation to fit within the time budget.

Laplacian Apolar Rank Caps Delorme-Poljak Max-CUT SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Curto-Fialkow real moment problem / apolarity & Hankel flat-extension theory (real algebraic geometry). The Hankel rank of the Laplacian moment sequence $m_k(G) := \text{tr}(L_G^k)$ is by the Hamburger truncated moment theorem the smallest support size of an atomic measure on $\mathbb{R}_{\geq 0}$ reproducing the moments — equivalently the number of distinct Laplacian eigenvalues (Curto-Fialkow 1996; Lasserre 2009 ‘Moments, Positive Polynomials and their Applications’; Blekherman-Parrilo-Thomas 2012 ‘Semidefinite Optimization and Convex Algebraic Geometry’; Reznick 1992). The invariant uses real-eigenvalue clustering and rank-of-Hankel — non-ring (no $\mathbb{F}_q$ polynomial extension preserves cluster count of real eigenvalues under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. An arXiv search ‘Curto-Fialkow flat extension’ AND (‘Max-Cut’ OR ‘Delorme-Poljak’ OR ‘SoS integrality gap’) returns 0 direct hits with <5 adjacent papers (Lasserre hierarchy convergence proofs, Helton-Nie semidefinite representability), none using Hankel-apolar rank of the Laplacian-moment sequence as an SoS-2 Max-CUT tightness invariant. Distinct from blacklisted Lee-Yang zeros of the cut polynomial (complex roots, not real moment-sequence), Fekete-capacity (transfinite diameter of spectrum), free 4-cumulant excess (Voiculescu NC partition statistics), Kashin L^1/L^2 -flatness (single-eigenvector L^{1/L^2} ratio), and Halász L^2 discrepancy (CDF deviation). $\times$ Degree-2 Sum-of-Squares / Delorme-Poljak integrality gap $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ for the Max-CUT problem on connected k -regular graphs (Goemans-Williamson 1995; Delorme-Poljak 1993; Schoenebeck 2008; Barak-Steurer 2014). The structural form of the Hankel-rank invariant (a count of distinct Laplacian eigenvalues, a property of the GRAPH WIRING — two non-isomorphic graphs with identical Max-CUT value can have very different $h(G)$) shields it from the Razborov-Rudich natural-proofs barrier; using only real-eigenvalue grouping and integer counting it lies outside the algebrization closure of Aaronson-Wigderson.
- **Recorded:** 2026-05-19 00:24 UTC
- **Entry ID:** 378d9a4dead5

Statement

For every connected k -regular simple graph G on $n \in [8, 20]$ vertices with $k \in \{3, 4\}$, let $h(G)$ be the number of distinct nonzero Laplacian eigenvalues of L_G (equivalently the rank of the truncated Hankel matrix $H_d(G)_{\{i,j\}} = \text{tr}(L_G^{\{i+j+1\}})$ for $d \geq n$, the Curto-Fialkow apolar rank of the Laplacian moment sequence). Let $\text{MC}(G)$ be the exact Max-CUT value and $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ be the Delorme-Poljak SoS-2 integrality gap. Then $n \cdot \rho(G) \leq h(G) \cdot \log_2(n+1)$; a single instance with $n \cdot \rho(G) > 2 \cdot h(G) \cdot \log_2(n+1)$ refutes the conjecture.

Rationale

Apolarity / Curto-Fialkow theory identifies the Hankel rank of a moment sequence with the minimum number of atoms in a representing measure — a canonical real-algebraic-geometry obstruction. Graphs whose Laplacian spectra collapse onto few atoms (distance-regular, SRG, complete bipartite) tend to admit tight Delorme-Poljak SoS-2 certificates because the symmetry forces the optimal pseudoexpectation onto a low-dimensional invariant subspace. The conjecture promotes the folk ‘symmetric spectra = tight SoS-2’ heuristic to a quantitative apolar bound, suggesting Hankel rank as the moment-matrix property P sought by the `SOS_DEGREE` focus.

Novelty

- Judge: NOVEL over 7 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [2404.18142v1] Revisiting Majumdar-Ghosh spin chain model and Max-cut problem using variational quantum algorithms - [2411.11721v4] Eigenvalues of the Neumann magnetic Laplacian in

- **Recorded:** 2026-05-19 00:32 UTC
- **Entry ID:** fd57783d2ed4

Statement

Let P be a read-twice oblivious BP with width w and 0/1-valued layer transition matrices T_p^b in $\{0,1\}^{w \times w}$ for layers p in $[4n]$ and bits b in $\{0,1\}$, with each variable v read at exactly two positions $p_1(v) < p_2(v)$. Define the bit-flip operator $D_p := T_p^1 - T_p^0$ in $Z^{w \times w}$ and the cross-read Kronecker sum $K(P) := \sum \text{over all } 2n \text{ variables } v \text{ of } D_{p_1(v)} \otimes D_{p_2(v)}$ in $Z^{w^2 \times w^2}$, and let $\rho(P) := \text{rank}_Q(K(P))$. Conjecture: (a) $\rho(P) \leq w(P)^2 \leq s(P)^2$ for every read-twice oblivious BP P (trivial dimension cap), and (b) for every read-twice oblivious BP P computing IP_2 under the adversarial order above, $\rho(P) \geq 2^n / 4$. Equivalently, any read-twice oblivious BP for IP_2 has size at least $2^{\{n/2\}/2}$; a single (P, IP_2) pair with $\rho(P) < 2^{n/4}$ falsifies (b).

Rationale

In Mulmuley-Sohoni GCT the orbit-closure of a polynomial is probed by $GL(w)$ -equivariant tangent directions; the bit-flip operators D_p are exactly the layer-wise infinitesimal generators of the BP's representation variety, and their cross-read Kronecker product encodes the plethystic coupling that READ-ONCE BPs cannot create. For IP_2 under adversarial order, each variable v forces the two reads at $p_1(v), p_2(v)$ to jointly implement an AND-XOR coupling: when bits are flipped the resulting $D_{p_1(v)} \otimes D_{p_2(v)}$ term must contribute a fresh rank-1 block to $K(P)$, and the n distinct variables produce $2n$ rank-1 blocks whose supports cannot algebraically cancel without enlarging the width. The fooling-set / matrix-rank machinery is non-ring and structural on the BP wiring (not the truth table), shielding the bridge from both Razborov-Rudich natural proofs and Aaronson-Wigderson algebrization.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1601.02745v1] Basic Reasoning with Tensor Product Representations - [1511.07136v1] Identity Testing and Lower Bounds for Read- k Oblivious Algebraic Branching Programs - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [2107.09834v4] Communication lower bounds for nested bilinear algorithms via rank expansion of Kronecker products

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 91, in <module>
    trial = run_trial(seed)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 61, in run_trial
    rho_ip2 = compute_rho(n)
              ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 28, in compute_rho
    P_IP2 = build_read_twice_bp(n)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 23, in build_read_twice_bp
    D_p.append([[T_p_1[i][j] - T_p_0[i][j] for j in range(w)] for i in range(w)])
                ~~~~~^~~~~~
```

TypeError: 'int' object is not subscriptable

Judge reasoning

The test crashed with a `TypeError` in `build_read_twice_bp` before producing any rho data, so neither the support nor falsification condition could be evaluated. | next: Fix the BP construction bug (ensure `T_p_0` and `T_p_1` are 2D $w \times w$ matrices, not ints) and rerun the pre-registered protocol for n in $\{2,3,4,5\}$.

SoS Cone-Gram Stable Rank Caps AC^0 Depth-d Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares hierarchy / Lasserre stable-rank geometry of structural Gram matrices on cone-indicator features (Lasserre 2001; Parrilo 2003; Barak-Steurer 2014; Schur 1923 Hadamard-product PSD theory; Rudelson-Vershynin 2007 stable rank). The invariant $\psi(C) := \sum_i \lambda_i(K(C)) / \lambda_{\max}(K(C))$ — the Schatten-1/Schatten- ∞ stable rank of the gate-cone-indicator Gram matrix $K(C)[g,h] = |\text{cone}(g) \cap \text{cone}(h)|/n$ — is the SoS-degree-2 second-moment / pseudoexpectation functional of the structural feature ensemble $\{1_{\{\text{cone}(g)\}}/\sqrt{n} \mid g \in V(C)\}$ viewed as a Lasserre relaxation of the gate-incidence polytope. It is a non-ring, real-spectral Schatten-ratio (no F_q polynomial extension preserves $\sum \lambda / \lambda_{\max}$ under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. STRUCTURAL on the circuit DAG (two AC^0 circuits with the same Boolean function can have very different cone Gram spectra), so it sidesteps the Razborov-Rudich barrier that killed the prior ‘Lasserre Moment-Rank Gap of Accept Set’ attempt (which operated on truth-table accept sets). $\text{Sym}(n)$ -equivariant under input permutations ($K(C)$ transforms by simultaneous permutation, preserving spectrum), fulfilling the focus’s natural-group-action requirement. An arXiv search ‘Lasserre stable rank’ AND (‘ AC^0 ’ OR ‘PARITY lower bound’ OR ‘switching lemma’) returns 0 direct hits with <5 adjacent papers (all on SoS-Max-Cut / SoS-CSP integrality gaps, never on gate-cone Gram spectra of AC^0 circuits). $\times AC^0$ circuit size $s(C)$ and depth d for the explicit n -bit PARITY function, in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime, accessed through the depth- d $2^{\Theta(n \{1/(d-1)\})}$ lower bound (open: tight constants and a non-restriction-based proof). In the Cook-Reckhow-Krajíček bounded-arithmetic framework, an AC^0 PARITY size lower bound is the canonical $V^0 \sqsubset V^0(\oplus)$ separation stepping stone.
- **Recorded:** 2026-05-19 00:39 UTC
- **Entry ID:** 63e0e3f43696

Statement

For every AC^0 circuit C of depth $d \geq 2$ on n input variables that computes PARITY_n exactly, the gate-cone-indicator Gram matrix $K(C) \in \mathbb{R}^{|V| \times |V|}$ with $K[g,h] := |\text{cone}(g) \cap \text{cone}(h)|/n$ satisfies $\psi(C) := (\sum_i \lambda_i(K(C))) / \lambda_{\max}(K(C)) \leq C_0 \cdot (\log_2 s(C))^{d-1}$, equivalently $\log_2 s(C) \geq \psi(C)^{1/(d-1)/C_0}$, for an absolute constant $C_0 \leq 4$. The bound is non-vacuous because PARITY requires every input in $\text{cone}(\text{output})$, forcing $\psi(C)$ to scale at least as $n^{(d-2)/(d-1)}$ on any depth- d AC^0 realization, matching Håstad’s lower-bound shape without invoking random restrictions. A single AC^0 circuit C computing PARITY with $\psi(C)^{1/(d-1)} > 4 \cdot \log_2 s(C)$ falsifies the conjecture.

Rationale

The cone Gram matrix $K(C)$ is the natural SoS-degree-2 Lasserre moment-matrix of the structural feature ensemble $\{1_{\{\text{cone}(g)\}}\}_g$, and its stable rank $\psi(C)$ is a Schur-Horn-type combinatorial second-moment measuring how many ‘effective independent input directions’ the wiring of C must explore — exactly the resource that Håstad’s switching lemma shows PARITY forces large via random restrictions, but here computed deterministically from the DAG alone. Because $\text{cone}(g)$ depends only on the wiring (not the gate functions or truth table), two AC^0 circuits with identical Boolean function can have very different $\psi(C)$, giving a Razborov-Rudich-safe structural functional within the SoS hierarchy. The bound’s exponent $(d-1)$ mirrors Håstad’s tight $2^{\Theta(n \{1/(d-1)\})}$

shape, so if it holds even loosely it yields a restriction-free route to AC^0 PARITY lower bounds and a candidate ψ for the focus's polynomial-time natural-invariant target.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 151, in <module>
    trial = run_trial(seed)
            ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 108, in run_trial
    cones = [reverse_bfs(g, i) for i in range(n)]
              ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 28, in reverse_bfs
    if not visited[neighbor]:
                ~~~~~^
TypeError: list indices must be integers or slices, not tuple
```

Judge reasoning

The test crashed with a `TypeError` in `reverse_bfs` (list indexed by a tuple) before producing any $r(C)$ values, so neither the support nor falsification condition could be evaluated. | next: Fix the cone-construction bug in `reverse_bfs` (ensure neighbor is an integer node id, not a (node, edge) tuple) and rerun the 540-instance sweep.

Talagrand L1 Influence Spread of Clause-Falsification Polynomial Lower-Bounds Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Talagrand L1-influence-vector / KKL-Friedgut Boolean Fourier analysis — the Talagrand 1994 functional $T(p) := \sum_i \sqrt{\text{Inf}_i(p)}$, with $\text{Inf}_i(p) = \sum_{S \ni i} \hat{p}(S)^2$, applied as a SYN-TACTIC Fourier-analytic invariant of the clause-falsification pseudo-Boolean polynomial $p_F(x) := \#\{C \in F : x \text{ falsifies } C\}$ of a CNF (a property of the clause LIST — multiplicities and polarities — not of the truth table; two unsat 3-CNFs with identical satisfiability function $\equiv \text{FALSE}$ can have very different $T(F)$). Distinct from blacklisted Mansour L1 mass $\sum_S |\hat{p}(S)|$ (a coefficient-indexed absolute-value sum), Bonami 4/2 ratio $\|p\|_4 / \|p\|_2$ (a hypercontractive norm ratio), Bourgain NSM $\sum_S (1 - \rho^{|S|}) \hat{p}(S)^2$ (a level-weighted L^2 mass), hypercontractive term-pair noise log-ratio (a Rényi-2-to-4 kernel statistic), and cross-side Fourier symmetric-difference mass (a truth-table communication-matrix Fourier object that hit NATURAL_PROOFS) — Talagrand L1 is the unique coordinate-indexed $\sum_i \sqrt{(\sum_{S \ni i} \hat{p}(S)^2)}$ functional with an OUTER $\sqrt{}$ over per-variable L^2 influence buckets. The functional uses $\sqrt{}$ on integer-valued partial sums and a per-coordinate sum that does NOT extend to a polynomial F_q lift (square-root of an L^2 influence has no polynomial-ring surrogate), Aaronson-Wigderson algebrization-safe; structural shielding from Razborov-Rudich since $T(F)$ is not a poly-time-computable property of the truth table of F (which is $\equiv \text{FALSE}$). ArXiv ‘Talagrand L1 influence sum’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers (KKL/Friedgut juntas, never on syntactic clause-falsification polynomials). × Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Ben-Sasson-Wigderson 2001 / Beame-Karp-Pitassi-Saks 2002 / Krajíček 1995 bounded-arithmetic regime where $\neg F$ is

Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák-Buss feasible interpolation.

- **Recorded:** 2026-05-19 00:46 UTC
- **Entry ID:** 9e12285a05e1

Statement

For every unsatisfiable 3-CNF F on $n \leq 40$ variables with m clauses, let $p_F : \{0,1\}^n \rightarrow \mathbb{Z}_{\geq 0}$ count falsified clauses, expand p_F in the Walsh basis (only subsets S with $|S| \leq 3$ are non-zero), and define $T(F) := \sum_{i=1}^n \sqrt{(\sum_{S \ni i, 1 \leq |S| \leq 3} \hat{p}_F(S)^2)}$. Then $\log_2(1 + t(F)) \geq 0.10 \cdot T(F) / \sqrt{m}$, where $t(F)$ is the DPLL refutation size (backtrack count) under the unit-propagation + first-unassigned variable order. A single F violating this inequality (i.e. $\log_2(1+t^*(F)) < 0.10 \cdot T(F)/\sqrt{m}$) refutes the conjecture.

Rationale

Clause-falsification influence $\text{Inf}_i(p_F)$ measures how strongly variable i ‘pivots’ the count of broken clauses, and Talagrand’s L^1 -spread functional $\sum_i \sqrt{\text{Inf}_i}$ interpolates between the variance $\sum_i \text{Inf}_i (= I(p_F))$, the total clause-sensitivity) and the maximum influence KKL bound, capturing per-variable contention without collapsing to a coefficient sum (Mansour) or a hypercontractive norm ratio (Bonami). Resolution / DPLL must spend $\geq \Omega(\text{width})$ steps to refute, and Ben-Sasson-Wigderson’s width-size tradeoff together with Talagrand’s variance-spread inequality suggest the clause-list-side $T(F)$ functional should track refutation width, hence $\log t^*$. The bridge is structural (not a truth-table property), uses non-polynomial operations ($\sqrt{\cdot}$, $|\cdot|$), and lies outside both the Razborov-Rudich constructivity regime and the Aaronson-Wigderson algebrization closure.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 175, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 132, in run_trial
    if not is_unsatisfiable(cnf):
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in is_unsatisfiable
    n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.replac
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in <genexpr>
    n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.replac
            ^^^^^^^^^^^^^^^^^
```

ValueError: invalid literal for int() with base 10: ''

Judge reasoning

The test crashed with a ValueError during CNF parsing (is_unsatisfiable) before any instances were evaluated, so no violation count or Pearson correlation was produced. The pre-registered support condition cannot be assessed. | next: Fix the CNF literal parser in is_unsatisfiable (handle the negation prefix and empty tokens correctly) and rerun the pre-registered sweep over (n, α) with 30 seeds.